Osamah Alananzeh

Computer Engineer | Backend Developer

+962795224589
osamahalananzeh@gmail.com
Linkedin
Github
Portfolio

Summary

Computer Engineering graduate and Backend/Systems Engineer with strong expertise in C/C++, networking, and concurrent programming. Experienced in building high-performance, low-level systems including HTTP servers, Unix shells, and multithreaded applications. Skilled in designing scalable backend architectures, REST APIs, and optimizing system performance. Hands-on experience with real-world projects involving networking, AI integration, and system-level debugging.

Education

BACHELOR'S DEGREE IN COMPUTER ENGINEERING
AlBalqa'a Applied University – Oct 2021 - Oct 2025

42 CORE CURRICULUM DIPLOMA IN COMPUTER SCIENCE
42 Amman – Aug 2024 - Aug 2026

Core Skills

- Systems Programming
- Backend Development
- Networking & Protocols
- Concurrency & Multithreading
- REST API Design & Integration
- Database Design
- Performance Optimization
- Debugging Complex Systems

Technical Skills

- **Programming Languages:** C, C++, C#, JavaScript
- **Backend & Networking:** REST APIs, HTTP/1.1, TCP/IP, Sockets, Client-Server Architecture
- **Core Computer Science:** Data Structures & Algorithms, OOP, Memory Management, System-Level Programming
- **Concurrency & Multithreading:** POSIX Threads (pthreads), Mutexes, Synchronization, Deadlock Prevention
- **Systems & Unix:** Processes, Signals, Pipes, File Descriptors, Environment Variables, Shell Redirections
- **Databases:** PostgreSQL, SQL Server
- **Frameworks:** .NET, React
- **Tools:** Git, GitHub, Valgrind, Helgrind, Makefiles, Postman, cURL, Siege, SQL Server Management Studio (SSMS), Visual Studio, VS Code , pgAdmin

Certifications

- Foundational C# with Microsoft , freeCodeCamp & Microsoft — Nov 28, 2025
- C# Level 1 , ProgrammingAdvices — Dec 26, 2025
- Database Level 1 - SQL (Concepts and Practice) , ProgrammingAdvices — Feb 1, 2026
- OOP As It Should Be In C# , ProgrammingAdvices — Feb 6, 2026
- Database - SQL (Projects & Practice) , ProgrammingAdvices — Feb 6, 2026
- C# & Database Connectivity , ProgrammingAdvices — Feb 14, 2026
- Shadow Code (Problem Solving Track) , IEEE BAU Computer Society — 1st Place, May 18, 2025
- Docker Training Course for the Absolute Beginner , KodeKloud — Mar 31, 2026

Projects

Minishell (C)

- Developed a mini Unix shell supporting pipes, redirections, environment variables, and process management.
- Improved modularity and debugging using Valgrind and Makefiles.

Cub3D (C, MiniLibX)

- Implemented a 3D rendering engine using raycasting with textures, collisions, and real-time player movement.

Advanced C++ Projects (OOP, STL, Templates) — 42 School

- Completed C++ modules covering OOP, memory management, inheritance, polymorphism, templates, and exception handling.
- Built multiple projects emphasizing clean architecture, SOLID principles, and effective STL usage.

Graduation Project — Conversational Firewall Management: Chatbot-Guided Firewall Operations

- Developed an AI-powered conversational interface for firewall management using FastAPI, React, and vLLM.
- Interprets natural language to configure and monitor an OPNsense firewall via secure REST APIs, reducing human error.

Dining Philosophers – Multithreaded Systems Project

- Developed a multithreaded Dining Philosophers simulation in C using mutexes for synchronization, preventing race conditions and deadlocks.
- Implemented precise time management, thread-safe logging, and strict memory handling, with extensive testing for starvation and concurrency edge cases.

Webserv (C++, HTTP Server)

Developed a fully functional HTTP/1.0 web server in C++ handling multiple concurrent clients.

- Implemented request parsing, routing, CGI execution, and static file serving
- Managed concurrent connections using non-blocking I/O (poll)
- Designed configurable server architecture inspired by Nginx
- Validated performance and stability using cURL and Siege under concurrent load

Languages

ARABIC: NATIVE
ENGLISH: ADVANCED